

Budi Muhamad Abdurahman.

Cryptography and Public Key Infrastructure Engineer

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Profile

Results-oriented with a proven track record of working collaboratively with team members to achieve goals. Enjoys learning new things especially related to Information Technology and has experience in the field of Software Development for more than 5 years as well as experience working as a Cryptography and Public Key Infrastructure Engineer for more than 4 years.



Work Experience

Public Key Infrastructure Engineer

eMudhra Indonesia (Full-Time)

September 2025 to August 2026

Migration from EJBCA to emCA at Peruri CA (September 2025 - May 2026)

- Install the CA, TSA, OCSP, and eSign services, as well as any other required supporting components, in the development, staging, and production environments.
- Documentation of the installation, configuration, and troubleshooting processes for systems that have already been installed.
- Conduct stress tests, load tests, and UAT on the installed systems to ensure that all systems are functioning properly.
- Perform data migration from EJBCA to emCA, monitor the system after migration, and ensure that the entire system is functioning properly following migration.
- Providing consultation and training sessions on the Public Key Infrastructure system to Peruri CA's personnel.

Implementation of Certificate Lifecycle Management (CLM) at Bank Danamon (November 2025 - Juni 2026)

- Install a private CA, the CLM system, and other supporting tools in the development, UAT/staging, and production environments
- Integrate the ticketing system (JIRA), CLM, and Privileged Access Management (CyberArk) to automate the certificate issuance and deployment process.
- Ensure that the issued SSL certificates can be installed on various target servers, such as IIS, Apache, Nginx, ADCS, firewalls, and appliance-based target servers.
- Troubleshooting several issues and conducting consultation and training sessions on the Public Key Infrastructure (PKI) system for Bank Danamon team.
- Documentation of the installation, configuration, and troubleshooting processes for systems that have already been installed.

Migration from EJBCA to emCA at ezSign CA (April 2026 - June 2026)

- Install the CA, TSA, OCSP, and eSign services, as well as any other required supporting components, in the development, staging, and production environments.
- Documentation of the installation, configuration, and troubleshooting processes for systems that have already been installed.
- Perform data migration from EJBCA to emCA, monitor the system after migration, and ensure that the entire system is functioning properly following migration.
- Troubleshooting several issues and conducting consultation and training sessions on the Public Key Infrastructure (PKI) system for ezSign CA team.

Set up on-premises systems and installations for proof-of-concept (POC) purposes at several clients and system integrators, such as BSI, BTN, Bank Permata, Helios, and Synnex Metrodata Indonesia

Public Key Infrastructure and HSM Engineer

Akulaku - iTruzz (Full-Time)

August 2023 to September 2025

- Implementing the best security tools in iTruzz such as Identity Access Management (IAM), System Information and Event Management (SIEM), Vulnerability Assessment Scanner, Intrusion Detection System (IDS) and Secure Software Development Lifecycle (SSDLC).
- Built the system infrastructure in the Data Center and Disaster Recovery Center with the team.
- Implement and participate in ISO 27001:2022 audit as Information Security Management System (ISMS).
- Ensured smooth functioning of the certificate lifecycle, including Certificate Creation, Certificate Issuance, Publication Process, Renewal Process, and Revocation Process, in accordance with the regulations set forth by the Ministry of Communication and Information acting as RootCA Indonesia
- Integrated EJBCA-CA with HSM, EJBCA-VA, and Signserver devices
- Operate and maintain HSM Utimaco
- Developing Crypto Key Management System (CKMS) application using Java programming language integrated with Utimaco HSM as personal data encryption

Public Key Infrastructure and HSM Engineer

Digisign.id (Full-Time)

January 2021 to August 2023

- Studied and implemented several cryptography algorithms, such as Rivest Shamir Alderman (RSA) and Elliptic Curve Cryptography (ECC).
- Developed Key Management System (KMS) application using GOLANG programming language as the backend.
- Conducted development on SoftHSM and EJBCA, then combined them into a single docker image
- Learned EJBCA technology and created a blueprint for the development of a Certificate Lifecycle Management (CLM/CMS) application that will be created by Digisign.
- Performed testing and development on OCSP Responder.
- Ensured smooth functioning of the certificate lifecycle, including Certificate Creation, Certificate Issuance, Publication Process, Renewal Process, and Revocation Process, in accordance with the regulations set forth by the Ministry of Communication and Information acting as RootCA Indonesia.
- Ensured that the Certificate Revocation List (CRL) was always updated daily and checked the validity of certificates through the Online Certificate Status Protocol (OCSP).
- Performed repairs (error handling) if there were system failures in the EJBCA applications, HSM devices, HSK applications, and KMS applications.
- Archived information related to certificates such as certificate requests and revocations
- Stored and reported audit trails of registration, issuance, renewal, and revocation of certificates within 12 months of transactions.
- Performed Remote File System (RFS) Installation on Ubuntu server as an HSM client for remote communication with Hardware Security Module (HSM) nShield.
- Installed and configured software on the server such as Enterprise Java Bean Certificate Authority (EJBCA) as a Certificate Authority (CA), Enterprise Java Bean Certificate Authority (EJBCA) as a Validation Authority (VA) for Wildfly/JBoss, Signserver.
- Integrated EJBCA-CA with HSM, EJBCA-VA, and Signserver devices
- Generated Key Pairs (Key Generation Ceremony) for Digisign's root key through the EJBCA application
- Created several Digisign Certificate Profile Roots in the EJBCA-CA application using the RSA algorithm and a Bit length of 4096
- Protected key pairs using Administration Card Set (ACS) with a quantity of 5/10 and Operation Card Set (OCS) with a quantity of 2/4 on the HSM device

- Backed up the Digisign keys that have been generated using a FIPS-140 Level 3 flashdisk from the HSM device
- Installed High Scalability Keystore (HSK) software from nShield on Ubuntu server to request keypair generation
- Replaced regular hard disks on the Data Center (DC) server and the Disaster Recovery Center (DRC)
- Migrated Certificate Authority (CA) key to the new Administration Card Set (ACS) key protection to enable logging system on the HSM device
- Implemented mass revocation of over 600.000 user certificates (mass revoked) in June 2022 and migrated them to EJBCA with a New Certificate Profile

Full-Stack Developer

Fibr Corp (Full-Time)

April 2020 to January 2021

- Created web application to build mobile applications via the web using angular framework
- Managed Fibr.io application data using Firebase
- Conducted stress testing for Fibr.io application based on specific user intervals
- Developed backend on the application to automate the building process
- Built Fibr.io application and deployed it to the Play Store
- Changed login page application by implementing OTP
- Integrated Popaket API with Backend for delivery service



Education

Hacktiv8 Indonesia

October 2020 to January 2021 | Full-Stack Javascript Development

Entrust

June 2023 | nShield Certified System Engineer (nCSE)

Diponegoro University

2012 to 2018 | Bachelor of Science in Physics



Relevant Skills

- Key Management/Certificate Management
- Cryptography
- Hardware Security Module (HSM) from nShield and Utimaco
- RHEL/Ubuntu
- Enterprise Java Bean Certificate Authority (EJBCA)
- JBoss
- Node.Js Programming Language
- Golang Programming Language
- Flutter Programming Language
- Angular Javascript
- React Javascript
- Firebase
- Apache Tomcat
- Nginx
- Databases MySQL and Postgresql